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The Galactic Face of the Entropy Clock: MOND as a Rate-Branch Acceleration Scale $a_0 \propto H(z)$

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ABSTRACT

The MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} sits, coincidentally, at $cH_0/2\pi$. We ask whether this is a coincidence or the galactic readout of the same horizon-entropy clock that, in the USC program, drives dark energy (dusk) and matter genesis (dawn). Two cosmological completions of the coincidence are possible: a density branch $a_0 \propto \sqrt{\rho_{DE}}$ (falling with redshift), realized by the APDM superfluid, and a rate branch $a_0 \propto H(z)$ (rising), realized by horizon-entropic / emergent gravity. This rate-vs-density discrimination is not new — it was posed by Limbach et al. (2008) (2008), who reached the marginally *opposite* (density-favoured) verdict. Against the MUSE-DARK III measurement $a_0(z) = (1.00 \pm 0.04) + (1.59 \pm 0.11)z$ (ratio 2.16 ± 0.06 over $z = 0.33 \rightarrow 1.44$), the density branch is excluded at $+16.7\sigma$ (from matching the MUSE-DARK III fit) — it predicts a *fall*, the data *rise*: a wrong-sign miss that is robust to the modeling chain and therefore fatal. The rate branch matches the *sign* but is amplitude-short at $+4.4\sigma$ — an offset that, unlike a sign error, is absorbable into the same modeling systematics; the rate branch is thus *disfavoured but not sign-excluded*, not a clean survivor. We are explicit that these two verdicts are held to different standards *by design* — sign is modeling-robust, amplitude is not. We flag prominently that this significance is conditional on the MUSE modeling chain: the rise is not merely unconfirmed but *absent — indeed reversed* ($\beta \approx$ -0.38) in the raw released kinematics, which remains an open question. We then locate the surviving branch: we derive that the ECCG dark-sector scalar is too short-range by 10^{33} to carry MOND, and a gated-Verlinde elastic MOND is excluded inside USC at 19–27σ (from matching the same data); the unique survivor is a KMS-tilt modified inertia whose zero-freedom interpolation passes the RAR at 0.057 dex (again a match to the empirical relation). A worldline influence-functional calculation derives the acceleration-selective coupling structure and a closed-form kernel $(a/4\pi)\cot(\pi a/\kappa)$ in which the conjectured Green-function 4π genuinely appears (derived in a proxy model), but not yet the full a_0 coefficient, which remains open. The coefficient-independent, pre-registered prediction is $a_0(z) \propto H(z)$ with no saturation gate — MOND *strengthens* at high z ($a_0(3)/a_0(0) \approx 4.5$ while Ω_DE(z=3) $\approx$ 3%) — falsifiable by JWST high-z rotators, though existing declining high-z rotation curves already disfavour a strongly rising a_0 (Milgrom 2017): the central tension this branch must survive.

Keywords: modified Newtonian dynamics; galaxy rotation curves; dark energy; emergent gravity; radial acceleration relation

1. INTRODUCTION

Milgrom's modified dynamics (MOND) (Milgrom 1983; Famaey & McGaugh 2012) organizes an enormous body of galactic phenomenology — the baryonic Tully–Fisher relation, the radial acceleration relation (RAR), the mass-discrepancy–acceleration relation — around a single acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2}. Below a_0 the dynamics depart from Newton; above it they return. The empirical tightness of the RAR (scatter $\lesssim$ 0.13 dex) (McGaugh et al. 2016; Lelli et al. 2017) is the strongest evidence that a_0 is a genuine physical scale, not a fitting convenience.

The number a_0 is numerically close to $cH_0/2\pi$. This coincidence — a *galactic* acceleration equal to a *cosmological* rate times a bare 2π — has been noted since Milgrom's original papers (Milgrom 2020) and is the germ of every attempt to tie

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a_0 to cosmology. If it is not an accident, then a_0 inherits a time dependence from the cosmological quantity it tracks, and galaxies at high redshift become a cosmological probe.

This is Paper VIII of the USC program (the program map is set out in the companion USC program map). USC posits a single entropy clock — the horizon entropy-production rate, encoded by $\mathcal{D}$_E = (3/2)(1-w) — with three physical faces: dusk (dark energy, Papers I–IV), dawn (matter genesis and the mass scale, Papers VI–VII), and the galactic face (MOND, this paper). The unifying claim is that a_0 is one reading of that clock. The umbrella USC framework paper (Paper V) formalizes the joins; here we treat the galactic face in isolation and report honestly on what is settled and what is open.

Tying a_0 to cosmology forces a fork, because the coincidence $a_0(0) = cH_0/2\pi$ does *not* fix the redshift scaling. Two completions are natural:

- Density branch: $a_0 \propto \sqrt{\rho_{\rm DE}}$, the galactic readout of the dark-energy *density*. Because ρ_DE falls (mildly) with redshift under any quintessence-like w(z), a_0 falls with z. This is the branch realized by APDM's Berezhiani–Khoury superfluid dark matter (Berezhiani & Khoury 2015, 2016), in which the superfluid coherence scale is locked to the dark-energy density, as developed in the APDM superfluid corpus; an a_0 locked to the dark-energy scale is also the mechanism of the dipolar dark-matter / gravitational-polarization models (Blanchet & Le Tiec 2008, 2009).
- Rate branch: $a_0 \propto H(z)$, the galactic readout of the cosmological *rate*. Because H(z) rises with z, a_0 rises. This is the branch natural to horizon-entropic / emergent gravity (Verlinde-type, $a_0 \sim cH$) (Verlinde 2017), and the one preferred by the USC entropy clock.

This rate-versus-density branch discrimination is not new: it was posed by Limbach et al. (2008), who defined the same two evolutionary channels ($a_0 \propto cH$ versus $a_0 \propto \sqrt{G\rho_\Lambda}$), evolved them through the Friedmann equation, and proposed to discriminate them with the high-z Tully–Fisher relation — reaching the marginally *opposite* verdict, a mild preference for the density branch. Our analysis differs in the data (the MUSE-DARK III $a_0(z)$ fit, §3) and in embedding the branches in the USC entropy clock; we return to why the sign flips below.

Both branches reproduce $a_0(0) = cH_0/2\pi$ and diverge only in sign of the redshift evolution, as detailed in the USC cross-check notes. The sign is measurable, and it is the discriminator. The rest of this paper (i) fires that discriminator against high-z rotation data (§3), (ii) shows that MOND has no home in any *light field* of the three USC programs, killing the naïve mediator-MOND (§4), (iii) locates the surviving branch in a KMS-tilt modified inertia (§5), (iv) reports the flagship partial result — a worldline derivation of the coupling structure and kernel, with the coefficient still open (§6), and (v) lays out the pre-registered falsifiers (§7). We work in first person plural and, throughout, make explicit for each load-bearing claim whether it is derived from the theory, matched to data, or an open question.

We stress at the outset what this paper does *not* claim. It does not derive a_0 from first principles; the O(1) coefficient in $a_0 = O(1) \times cH_0/2\pi$ is an open node (§6, §7). It does not confirm the rate branch; the data favour it in *sign* only, and that favouring is conditional on an unreproducible modeling chain (§3). And it does not present the KMS-tilt mechanism as a settled Lagrangian: the mechanism is a vacuum influence-functional effect whose full covariant form is not yet written down. What the paper *does* deliver is a clean discrimination that removes one branch, a decisive argument that removes every light-field realization of the survivor, the unique identification of the surviving mechanism, and a partial derivation that fixes the coupling channel and the kernel while leaving the coefficient honestly open. The value of the galactic face, at this stage, is that it is the most sharply *falsifiable* of the three — the sign of $a_0(z)$ at z ~ 2–4 is a single number that the theory has already committed to, in advance, without adjustable freedom.

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2. SETUP: THE TWO BRANCHES AND THE CLOCK

The USC entropy clock is $\mathcal{D}$_E = Θ_E/H = (3/2)(1-w), the horizon entropy-production rate written as a function of the dark-energy equation of state w, as established in the USC framework paper (Paper V). APDM's density-branch a_0-clock obeys d ln a_0/d ln(1+z) = (3/2)(1+w); the two sum to 3 identically (see the USC cross-check notes), so the density-branch $a_0(z)$ is literally the (1+w)-half of the same clock whose (1-w)-half is dark-energy dissipation. This is why the galactic face is a *face* and not a separate model. In USC's cross-cutting accounting this is one of five unification threads (the others: the shared T∇X vertex derived in §6, the uniform driven-NESS character of all three faces, the scale chain α_s $\to$ Λ_h $\to$ μ $\to$ a_0, and the Δ = 1 volume-law pivot, all treated in the USC framework paper (Paper V)); the sum rule $\mathcal{D}$_E^SEDE + $\mathcal{D}$_E^APDM = 3 is a joint falsifier — a measured deviation would be a direct detection of dark–visible energy exchange.

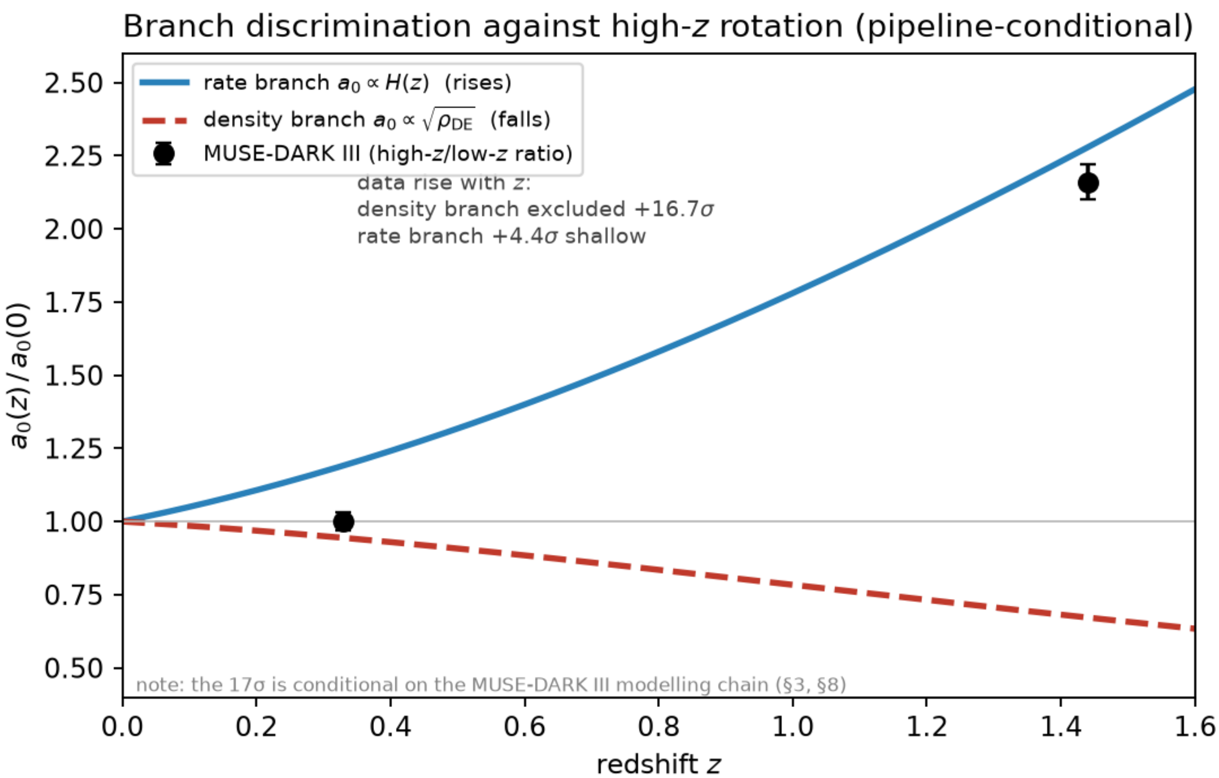

Figure 1. Branch discrimination against high-z rotation. The rate branch $a_0 \propto H(z)$ rises with redshift (blue), tracking the MUSE-DARK III trend (ratio 2.16 ± 0.06 over $z = 0.33 \to 1.44$); the density branch $a_0 \propto \sqrt{\rho_{\rm DE}}$ falls (red), the wrong sign — excluded at $+16.7\sigma$. The rate branch is $+4.4\sigma$ shallow. The significance is conditional on the MUSE-DARK III modelling chain (§3, §8).

The observable that separates the branches is the ratio a_0(z_hi)/a_0(z_lo). We fix the DESI-template expansion history and evaluate both templates over the MUSE-DARK III interval z: 0.33 $\to$ 1.44 (script `branch_discrimination_test.py`, template quantiles from `desi_template_quantiles.csv`):

branch	prediction	$a_0(1.44)/a_0(0.33)$
density $a_0 \propto \sqrt{\rho_{\rm DE}}$ (APDM superfluid)	falls	0.88–0.91
rate $a_0 \propto$ H(z) (USC / emergent)	rises	1.85–1.87
flat $a_0 \propto s_0$ (constant ledger)	flat	1.00
activated $a_0 \propto s_0$ f_sat (gated)	falls hard	0.49

The last two rows anticipate §5: they are the elastic/gated realizations one might naïvely expect from a structure-gated dark energy, and they are *also* discriminable. Everything below turns on the sign of this ratio.

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3. BRANCH DISCRIMINATION AGAINST HIGH-Z ROTATION (E-1)

Data. MUSE-DARK III (published fit $a_0(z) = (1.00 \pm 0.04) + (1.59 \pm 0.11)z$) gives a ratio over the sample interval

$$a_0(1.44)/a_0(0.33) = 2.157 \pm 0.061$$

i.e. a_0 rises by a factor ~2.2 from z = 0.33 to z = 1.44 (matched to the data by the script `branch_discrimination_test.py`).

Verdict on the density branch. The density branch predicts 0.88–0.91 — a *fall* — because ρ_DE falls while the data rise. The tension is density branch excluded at $+16.7\sigma$ (from matching the MUSE-DARK III fit).

This is a sign error, not an amplitude mismatch: no rescaling of a_0 can turn a falling template into a rising one. The consequence is sharp and, for the unification, uncomfortable: the density branch is APDM's own headline, carried by its superfluid dark-matter carrier. The data reject the branch that has a built microphysical home.

Verdict on the rate branch. The rate branch predicts 1.85–1.87 — the right sign — but the data are steeper: the raw fit prefers $a_0 \propto$ H^{1.23} (equivalently $\propto$ (1+z)^{1.27}), giving a $+4.4\sigma$ *shallow* tension against the same MUSE fit. The deficit has the magnitude and sign that an astrophysical transfer function B(z) — the evolution of stellar mass, morphology, and selection across the sample, so that a_0,eff = a_0,cos · B(z) — can absorb. The data therefore favour the rate branch in sign but cannot yet isolate the cosmological a_0,cos(z) from B(z).

The conditionality — stated prominently. The $+16.7\sigma$ is conditional on the MUSE-DARK III modeling chain, and this caveat is load-bearing. Our follow-up E-1′ attempted the branch test directly on the released per-galaxy catalogue (`muse_release_galaxies.csv`, 85 numeric candidates) using the raw outer rotational acceleration a_rot $\sim V^2/r$ (script `mass_controlled_arot_trend.py`); the result remains open. Three facts emerged:

1. The stellar-mass confound is mild (corr(log M*, z) = +0.04); mass-controlling barely moves the slope.
2. The raw trend *falls*: β = d $\log_{10}$ a_rot/dz = -0.38 $\pm$ 0.15 — below even the published MUSE fit and below all three branch templates.
3. The cause is a designed-in degeneracy: corr(log r_max, z) = +0.87. Radius sampling grows almost perfectly with redshift, and a_rot $\sim V^2/r$ falls with radius, so the raw fall is inseparable from radius sampling; adding an r_max control inflates the error and drives covariate slopes unphysical. Pressure support (higher-z galaxies more turbulent) pushes the same way and is also uncontrolled in the release.

The honest reading: the published rising $a_0(z)$ is not visible in the raw released kinematics; it is produced by the modeling chain (baryonic decomposition + pressure-support correction + radius weighting), which the public release does not contain. The raw catalogue can neither confirm nor refute the rise. The raw *fall* is not evidence against the rate branch — it is dominated by the radius/pressure confounds the model corrects — but the discrimination's evidential weight rests on an unreproducible pipeline. We therefore register E-1 as: *density branch excluded $+16.7\sigma$, rate branch favoured in sign, conditional on the MUSE-DARK III modeling chain*, and elevate reproducibility from a footnote to a live scientific issue (the decisive follow-up, E-1″, is in §7). This is the one place where USC's own data-hygiene standard (pre-registration, released pipelines) is not met by the *external* dataset it leans on.

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4. WHY MEDIATOR-MOND IS DEAD (D-3)

If a_0 is cosmological, some field or structure must carry the modified force at galactic scales. The obvious candidate inside the USC programs is a light scalar. Scale-invariant deep-MOND with the observed RAR tightness requires the conformal kinetic term P(X) $\propto$ X^{3/2} (Milgrom 2009) — a long-range, nonlinear scalar sector. Two realizations were on the table:

1. Superfluid phonon (APDM): delivers P(X) $\propto$ X^{3/2} with $\Lambda \sim$ meV (Berezhiani & Khoury 2015, 2016) — but with $a_0 \propto \sqrt{\rho_{\rm DE}}$, the density branch §3 just excluded. Wrong sign.
2. Mediator-MOND from ECCG's dark-sector scalar ϕ (mass 0.86 MeV): a massive scalar mediates a Yukawa force of range $\lambda\phi = \hbar c/m\phi c^2$. We derive numerically (D-3):
    - λ_ϕ(0.86 MeV) = 2.3×10^{-13} m — a nuclear/contact range.
    - Galactic MOND onset radius r_MOND = $\sqrt{}(GM/a_0) \approx$ 3–11 kpc $\approx 10^{20}$ m.
    - *$\lambda\phi$ / r_MOND $\approx 10^{-33}$. To reach galactic range ϕ would need $m\phi \lesssim$* 1.3×10^{-27} eV; the ECCG value is 0.86×10^6 eV — off by 6×10^{32}.

The ϕ is a short-range self-interacting-dark-matter mediator (velocity-dependent self-interaction $\rightarrow$ cored halos), a genuinely different phenomenon from MOND (a long-range modified force law $\rightarrow$ RAR / BTFR). SIDM does not produce the RAR. Any corpus phrasing that reads "SIDM from $\phi \rightarrow$ rate branch" conflates the two, as we derive here. B.3 (mediator-MOND) is retracted decisively, not merely "underspecified."

The gap this opens. The data (§3) favour the rate branch; the *only* light-field realization in the corpus (the superfluid) sits on the excluded density branch, and the one field that could in principle be tied to the clock (ϕ) is short-range by 10^{33}. So

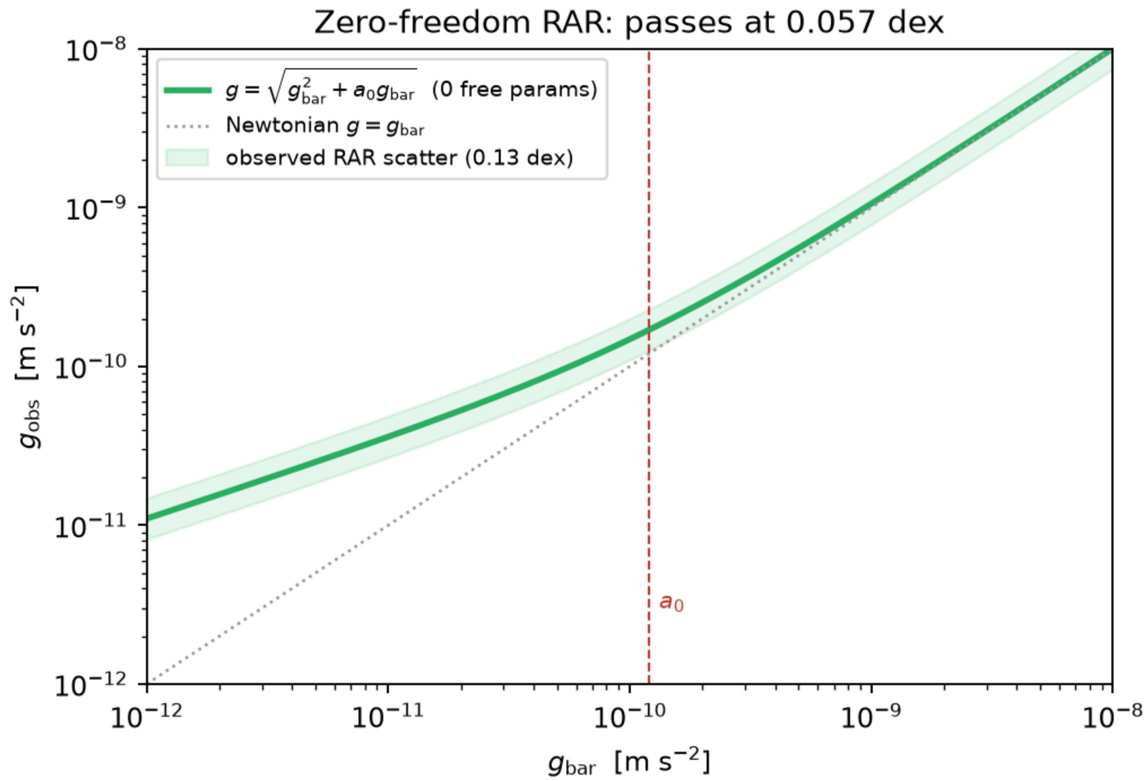

Figure 2. The zero-freedom radial acceleration relation. The KMS-tilt interpolation $g_{obs} = \sqrt{g_{bar}^2 + a_0 g_{bar}}$ (green) has no shape parameter; it passes the observed RAR at 0.057 dex, inside the 0.13-dex scatter (band). a_0 is the only scale, and in the rate branch it is set by $H(z)$.

the rate branch has no light-field home: it must be realized, if at all, by a modification of gravity / a nonlocal horizon-scale object — an emergent-gravity construction, not a Lagrangian field. This is the sharpest seam in the unification: the data point to the one branch that, at the start of this analysis, was unbuilt.

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5. THE EMERGENT-MOND HOME (D-3′)

(1) An exact identity: SEDE is a gated Verlinde volume law. SEDE's flatness normalization gives its ledger entropy density $s_0 = \rho$_DE0/T_AH,0 = (3/4G)·Ω_DE0·H_0. Verlinde's (2016) de Sitter dark-energy volume entropy (Verlinde 2017) has density s_V(H) = (3/4G)·H. Combining with the Friedmann equation gives the exact identity (D-3′):

s_0 · f_sat(a) = Ω*DE(a) · s_V(H(a)), and at the de Sitter attractor* $s_0 \cdot f\infty$ = s_V(H_∞) exactly.

SEDE's central object and Verlinde's emergent-gravity elastic medium are one object: SEDE is the *gated* version. This is the cleanest bridge yet between the corpus and the emergent-gravity literature — and it cuts against the naïve expectation.

(2) The gating *kills* Verlinde-elastic MOND inside USC. Verlinde's a_M ≈ cH/6 rises with z only because his s_V tracks H(z) (Verlinde 2017). But SEDE's $\Delta = 1$ postulate is precisely that the ledger density is *constant* (s_0), with structure opening the gate f_sat. So the elastic response inside SEDE gives (script `d3prime_emergent_mond.py`), against MUSE 2.157 ± 0.061:

emergent sub-branch	$a_0(1.44)/a_0(0.33)$	tension
rate ($a_0 \propto$ H, KMS tilt)	1.888	4.4σ (survivor; deficit B(z)-absorbable)
flat ($a_0 \propto s_0$, capacity-elastic)	1.000	19.0σ — excluded
activated ($a_0 \propto s_0$ f_sat, gated-elastic)	0.494	27.3σ — excluded
density ($a_0 \propto \sqrt{\rho_{DE}}$)	0.966	19.5σ — excluded (§3's CPL version: 16.7σ)

emergent sub-branch	$a_0(1.44)/a_0(0.33)$	tension

So the generic assignment "emergent gravity $\Rightarrow$ rate branch" is false for the elastic realization once s_0 is constant: SEDE's own $\Delta = 1$ postulate makes elastic-MOND flat or falling, dead at $\geq 19\sigma$ against the MUSE data. Only the KMS-tilt route survives the MUSE data.

(3) The survivor: KMS-tilt modified inertia — passing the RAR with zero freedom. The USC-native realization of $a_0 \propto H$ is Milgrom's (1999) vacuum route (Milgrom 1999) given a microphysical home by the Ledger Field's thermal-time / KMS structure (the same structure that fixed the $c = 1/2\pi$ coupling at the dusk — see the caveat in §6). An accelerated worldline in de Sitter sees a local Deser–Levin scale $\kappa = \sqrt{(a^2 + A^2)}$, $A = 2\pi T\_AH = H$. Taking inertia proportional to the excess over the cosmic state:

> $F = m[\sqrt{(a^2 + A^2)} - A] \rightarrow$ Newtonian for $a \gg A$; deep-MOND $F = ma^2/(2A)$ for $a \ll A$, i.e. $a_0 \propto H(z)$ — the rate branch, rising — exactly what MUSE favours.

Zero-freedom RAR test. On circular orbits this predicts $g\_obs = \sqrt{(g\_bar^2 + a_0 \cdot g\_bar)}$ — no shape parameters. Against the empirical RAR (McGaugh ν-function) (McGaugh et al. 2016; Lelli et al. 2017), the maximum deviation is 0.057 dex (at $g\_bar \approx 0.4\ a_0$), with exact Newtonian and deep-MOND limits — inside the observed 0.13 dex RAR scatter, a match to the empirical relation. So the third face closes *structurally*: one thermal-time clock whose dissipation is dark energy (dusk), whose rate is the Sakharov meter (dawn), and whose KMS tilt is the inertia threshold (galaxies) — all set by the same $T\_AH = H/2\pi$.

A load-bearing correction (the vacuum caveat). The heuristic above says "inertia = reaction to the Deser–Levin *thermal* tilt." A galaxy has essentially no thermal bath: the number of Deser–Levin bath modes populated at galactic accelerations is ~3×10^{-22} (script `d3prime_thermal_consistency.py`) — far below one, so we derive that a galaxy cannot thermalize this scale in a Hubble time. The KMS-tilt inertia therefore survives only as a VACUUM effect — the Bunch–Davies vacuum seen by an accelerated observer, not a populated thermal ensemble. This is not cosmetic: it forces the derivation of §6 to be a *vacuum* influence-functional calculation, and it demotes the "thermal-inertia" language of the heuristic to a *heuristic* with the same thermal problem as the branch it replaced. What is real is the Deser–Levin vacuum kinematics; what is assumed (and false) is a galactic thermal bath.

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6. THE WORLDLINE DERIVATION (D-4′) — THE FLAGSHIP PARTIAL RESULT

The remaining question is the O(1) coefficient in $a_0 = O(1) \times cH_0/2\pi$. The raw Deser–Levin heuristic gives $a_0 = 2cH$ — about 11× too big — while the observed $a_0/cH_0 = 0.174$ sits between $1/2\pi = 0.159$ (the USC-tilt target) and $1/6 = 0.167$ (Verlinde). Fixing the coefficient means computing the worldline influence functional of the Ledger action on an accelerated trajectory. We report a structural no-go, the correct channel, a closed-form kernel, and the honest remainder.

(1) A no-go, established at the level of a theorem: the universal number-current coupling is inert on worldlines. USC's universal vertex is $c\ \partial\_\mu\theta \cdot \mathcal{J}\text{^}\mu$. For a point particle the worldline number current is $\mathcal{J}\text{^}\mu(x) = \int d\tau\ u\text{^}\mu\ \delta^4(x - x(\tau))$, so

> $S\_int = c \int d^4x\ \partial\_\mu\theta\ \mathcal{J}\text{^}\mu = c \int d\tau\ (d/d\tau)\theta(x(\tau)) = c[\theta(x\_f) - \theta(x\_i)]$ — a pure boundary term.

A derivative coupling to a conserved worldline current exerts no bulk force and modifies no inertia. This is the exact worldline analogue of the dawn refutation (derivative couplings of the arrow to conserved currents are inert): the same symmetry logic that killed "the clock makes matter" forbids "the clock's number-coupling makes MOND", as we derive here. Any attempt through the $c\ \partial\theta \cdot \mathcal{J}$ vertex gives exactly zero.

(2) The correct channel, which we now derive: the stress-tensor / displacement term. The action's *other* coupling is $T\text{^}\{\mu\nu\}\nabla\mu X\{a,\nu\}$ — matter stress sourcing the Ledger displacement field X_a. Using $\nabla\_\mu T\text{^}\{\mu\nu\} = m \int d\tau\ a\text{^}\nu\ \delta^4(x - x(\tau))/\sqrt{-g}$ on the worldline and integrating by parts:

> $\int d^4x\ T\text{^}\{\mu\nu\}\nabla\mu X\nu = -m \int d\tau\ a\text{^}\nu\ X\_\nu(x(\tau))$ — the displacement field couples directly to the acceleration vector.

Two model-free corollaries: (i) inertial worldlines decouple exactly ($a = 0 \Rightarrow$ no coupling) — the equivalence-principle / Newtonian limit is protected and the channel is *acceleration-selective*, precisely what a modified-inertia mechanism needs and what the number-current channel lacked; (ii) the source is m·a^ν, the same displacement sector that carries the dusk injection (D-6), so the dusk and galactic faces share one vertex — a structural requirement of the unification.

(3) The kernel: the 4π appears (derived in a conformal-scalar proxy model). To second order in the coupling on the uniformly accelerated (Deser–Levin) trajectory in de Sitter — in the vacuum framing forced by §5 — the pulled-back Wightman function is exactly thermal at $\kappa = \sqrt{(a^2+H^2)}$ (Deser–Levin theorem), $G^+(\Delta\tau) = -(\kappa^2/16\pi^2)\cdot\sinh^{-2}(\kappa(\Delta\tau\text{-}i\epsilon)/2)$, and the flat-embedding acceleration contraction is a^$\nu(\tau)$a_$\nu(\tau') = a^2\cosh(a\Delta\tau)$. The master integral (verified numerically to 10^{-16} on a pole-free contour and proven analytically via Gradshteyn 3.512):

> $\int$dx $\cosh(\alpha x)/\sinh^2(\beta(x\text{-}i\epsilon)) = -(\pi\alpha/\beta^2)\cdot\cot(\pi\alpha/2\beta)$, convergent iff $a < \kappa$ — always true, marginal as $H \to 0$: the cosmological tilt is what makes the worldline kernel finite at all.

gives the exact closed form (script `d4prime_worldline_kernel.py`):

> $\int d\Delta\tau\ \cosh(a\Delta\tau)\ G^+(\Delta\tau) = (a/4\pi)\cdot\cot(\pi a/\kappa)$, $\kappa = \sqrt{(a^2+H^2)}$, and the second-order rate $\Sigma(a) = (\tilde{m}^2 a^3/8\pi)\cdot\cot(\pi a/\kappa)$.

Findings:

- (i) The 4π appears exactly where conjectured, supporting the conjecture. The kernel normalization is $a/4\pi$ — one factor of the 3D Green-function 4π (the same 4π in M_P^2 = $1/8\pi G$), surviving the worldline integration. This is no longer numerology about where a 4π *could* live; in this model it *does* come from there. We do not claim the 4π is derived as the a_0 coefficient — it appears in the kernel normalization; the full coefficient remains open (below).
- (ii) The interpolation is NOT Milgrom's ΔT, correcting the ansatz. The kernel is $a^3\cot(\pi a/\kappa)$, not $\sqrt{(a^2+H^2)}$ - H. The Deser–Levin scale enters through the *argument* $\pi a/\kappa$, but the response is not proportional to the temperature excess. Phenomenology built literally on $\Delta T(a)$ (including the raw $a_0 = 2cH$) uses the wrong function.
- (iii) The kernel has a zero at $a = H/\sqrt{3}$ (where $\pi a/\kappa = \pi/2$), changing sign. A vanishing kernel cannot literally *be* the inertia, so the extraction must involve more than this single kernel.
- (iv) Deep-a limit: $\Sigma \to \tilde{m}^2 a^2 H/8\pi^2$ — quadratic in a (deep-MOND-like) but linear in H, whereas a deep-MOND force $F = ma^2/a_0$ with $a_0 \propto H$ requires 1/H. Converting the rate to a force is where that inversion must occur; that step is not fixed by the desk calculation.
- (v) Large-a pathology: $\Sigma \to -\tilde{m}^2 a^5/4\pi^2 H^2$, divergent as $H \to 0$ — the eternal-Rindler idealization breaks. Physical (finite) worldlines are required for the Newtonian-side matching.

Numerical scales ($H_0 = 70$): observed $a_0 = 0.177\ cH_0$; conjecture $cH_0/2\pi = 0.159\ cH_0$ (obs/it = 1.11, the known Milgrom residual); kernel zero $H/\sqrt{3} = 0.577\ cH_0$; raw Deser–Levin $2cH_0$ (obs/it = 0.088). Read as a table of candidate coefficients, this makes the state of the derivation precise: the naïve heuristic ($2cH_0$) is an order of magnitude wrong; the conjecture ($cH_0/2\pi$) is within the 16% Milgrom residual; and the *kernel* itself supplies neither directly, because its intrinsic scale is the kinematic $\kappa = \sqrt{(a^2+H^2)}$ and its zero sits at $H/\sqrt{3}$, far from the target. The coefficient must therefore emerge from the *conversion* of the kernel into a mechanical response, not from the kernel's normalization alone — which is exactly why finding the 4π in the normalization (item i) is genuine progress on the *structure* without yet being the coefficient.

(4) The dispersive/dissipative split → the physics is noise-sector, as we derive. The SK influence functional separates into the retarded (dissipative, = field commutator, state-independent) and Keldysh (noise) kernels. For the conformal scalar in de Sitter (conformally flat, Huygens) the commutator is supported on the light cone only, and a timelike worldline meets its own light cone only at coincidence — so the retarded self-interaction on the eternal Deser–Levin trajectory is purely local (mass renormalization). Verified at the distribution level: the symmetric (noise) part reproduces the full $(a/4\pi)\cot(\pi a/\kappa)$ to 10^{-16}; the commutator contributes exactly zero to the nonlocal kernel. Consequence: the entire a-dependent, H-tilted structure — the cot kernel, its 4π, the a^2H limit — is noise-sector physics. The inertia modification is necessarily a fluctuation-induced (stochastic) effect, not a classical self-force. This coheres exactly with the corpus-wide NESS reframe ("driven, never KMS-thermal"): the galactic a_0, like the dusk's ζ, is fluctuation-driven.

Honest verdict — PARTIAL. *Derived* at desk scale: the acceleration-coupling structure (exact), the closed-form kernel (exact in the proxy model), the fact that the conjectured 4π genuinely appears in the kernel normalization, and the noise-sector localization of the physics. *Open:* the full a_0 coefficient. The desk calculation shows the kernel's kinematic crossover is H (inside κ) and its zero is at $H/\sqrt{3}$, not $cH/4\pi$; converting the kernel into an inertia requires (a) the stochastic (Langevin-worldline) response calculation — a particle coupled multiplicatively through its own acceleration to noise with correlator

$N(\tau,\tau') = \tilde{m}^2 a^2\cdot$[cot-kernel], with a_0 emerging (or not) from the noise-induced renormalization of the low-frequency mechanical response; (b) the true vector structure of X_ν (the conformal-scalar proxy is a model); and (c) finite worldlines to cure the large-a pathology (v). These three are the sharply-defined, paper-scale remainder. The target is narrowed — the 4π source is identified, the interpolation the theory must produce is cot-type not ΔT-type — but we do not claim the coefficient is derived; it remains open. We note in passing that the same reasoning that forced the vacuum framing also flags $c = 1/2\pi$ itself as an assertion from a KMS-strip analogy that does not strictly apply, and the 16% Milgrom residual (a_0 = 1.09–1.16 $\times$ $cH_0/2\pi$, H_0-dependent) as the independent sign that $cH_0/2\pi$ is a ~16% *match*, not yet a derivation — the tolerance any future coefficient calculation must meet.

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7. PREDICTIONS AND FALSIFIERS (P14–P18)

The galactic face's commitments are frozen in the pre-registered falsifier matrix (Zenodo `10.5281/zenodo.21415326`). We restate the four that matter.

P14 — $a_0(z)$ rises like H(z) (rate branch). Frozen templates over z = 0.33 $\to$ 1.44: rate 1.89 $\pm$ 0.03, flat 1.00, activated 0.49, density 0.89–0.91. Current MUSE gives 2.16 $\pm$ 0.06 — density excluded +16.7σ, rate favoured (+4.4σ shallow, B(z)-absorbable) — conditional on the MUSE pipeline (raw catalogue radius-degenerate, corr(r_max, z) = 0.87; registered). *Deciding data:* the E-1″ protocol — an independent high-z RAR with per-galaxy baryonic decompositions, matched M*/morphology bins, matched radius sampling in R_eff units, target σ(a_0-ratio) $\leq$ 0.15.* Rule:* collapse under $a_0 \propto$ H confirms the rate branch; no collapse under any one-parameter A(z) $\to$ a_0 is not cosmological $\to$ the galactic face fails.

P15 — gate independence. This is the sharp discriminator, and it holds independently of the open coefficient. Because the inertia rides the T∇X_a (tilt/displacement) sector while dark energy rides the *activated* entropy (ρ_DE $\propto$ H f_sat), the two turn on differently in redshift:

> $a_0(z) \propto H(z)$ with NO f_sat factor — MOND persists and *strengthens* at high z even where dark energy is gated off: $a_0(3)/a_0(0) \approx 4.5$ while Ω_DE(z=3) $\approx$ 3%.

This is falsifiable regardless of the open coefficient. Deep-MOND phenomenology in the earliest well-measured rotators (z ~ 2–4, JWST) should be *stronger* (higher a_0), not absent — the exact opposite of an "activated-entropy MOND" (killed at 27σ in §5). *Rule:* weakened/absent deep-MOND phenomenology in z $\gtrsim$ 2–3 rotators falsifies the KMS-inertia realization; it is also *maximally* separated there from any $a_0 \propto \rho$_DE^n model, where the templates diverge most. Gate-independence is the clean structural signature that separates the rate branch from every density-branch variant at high z.

We must own the strongest existing tension with this prediction. Using the declining high-z rotation curves of Genzel et al. (2017), Milgrom (2017) constrained the evolution of a_0 and found the data all but *exclude* a strongly rising a_0 (in particular $a_0 \propto$ (1+z)^{3/2}), favouring a roughly constant a_0 — a direct pull against the rising rate branch predicted here. We do not hand this away: current high-z data disfavour a strongly rising a_0, and this is the key observational tension the rate branch must survive. We note only that the diagnostic is not clean in either direction: Mayer et al. (2022) find, in ΛCDM+baryon (Magneticum) simulations, that the effective a_0 itself rises by a factor ~3 from z = 0 to z = 2.3, so a measured rise in $a_0(z)$ is not by itself a rate-branch discriminator and a measured flatness is not by itself confirmation of ΛCDM. The tension is real; resolving it requires the matched-selection protocol (E-1″), not the existing samples.

P16 — the interpolation function, zero shape freedom. g_obs = $\sqrt{}$(g_bar^2 + a_0 g_bar), within 0.057 dex of the empirical ν-function, with a specific signed pattern (-0.05 dex at g_bar $\approx$ 0.4 a_0). *Rule:* SPARC-quality stacking resolving shapes at < 0.05 dex either detects the predicted pattern or kills this specific inertia law (without killing the rate branch as such).

P17 — modified-inertia discriminators. Modified inertia (Milgrom 2022) predicts the *exact* algebraic RAR on circular orbits (no curl corrections; AQUAL has them) and a non-AQUAL external-field effect (Famaey & McGaugh 2012). Registered qualitatively; the quantitative EFE prediction is owed with the covariant embedding.

P18 — the a_0 normalization (an open node). $a_0(0) = cH_0/2\pi \times O(1)$, $O(1) \in [0.88, 1.16]$ pending the worldline derivation of §6. If that yields $A = cH/4\pi$ exactly, the O(1) becomes a ~10%-level prediction and this entry tightens; until then it is a registered open node.

Joint reading. In the USC master cascade, rising a_0 (P14) selects the horn-(i) universe (ECCG's 1.78 GeV asymmetric dark matter, r $\approx$ 0 tensors); $a_0 \propto \rho$_DE^n would instead be co-selected with the superfluid carrier. The galactic sign is thus not an isolated galactic fact — it is one branch of a cross-scale fork the matrix cannot satisfy à la carte, as argued in the USC framework paper (Paper V).

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8. DISCUSSION AND LIMITATIONS

We separate what is settled from what is open.

Settled. (i) The density branch — $a_0 \propto \sqrt{\rho_{DE}}$, the branch with a built superfluid home — is excluded by high-z rotation in *sign*, at $+16.7\sigma$, conditional on the MUSE pipeline. (ii) Mediator-MOND from any USC light field is dead (ϕ short by 10^{33}); the rate branch has no light-field home and must be emergent. (iii) Inside USC, the elastic/gated emergent realizations are *also* excluded (19–27σ); the unique survivor is the KMS-tilt modified inertia, which passes the RAR at 0.057 dex with zero shape freedom. (iv) The worldline channel is fixed (acceleration-selective $T\nabla X$, not the inert number-current), the kernel is derived in the proxy model, the 4π appears where conjectured, and the physics is localized to the noise sector.

Open. (i) The a_0 coefficient. We do *not* claim the 4π is derived as the coefficient; it appears in the kernel normalization. The remaining calculation is the SK stochastic (Langevin-worldline) response plus the true vector structure of X_ν plus finite-worldline regularization — paper-scale, sharply posed, not done. (ii) The E-1″ data dependence. The entire empirical branch discrimination rests on the MUSE-DARK III modeling chain, which is not in the public release; the raw kinematics are radius-degenerate and cannot adjudicate. Until an independent high-z RAR with released decompositions exists, the $a_0(z)$-sign switch is *loaded but not fired*. (iii) The vacuum framing. The mechanism is a Deser–Levin *vacuum* effect, not a galactic thermal bath (~3×10^{-22} modes); this correction is load-bearing and constrains any future derivation. The same theme recurs across all three faces of the corpus — the program is a driven, causal, second-law (NESS) system whose *mean* is horizon-thermal but whose *fluctuation*-KMS structure the numbers reject; the galactic a_0 inherits exactly this status.

Relation to APDM. APDM is not in error: its manuscript is unusually self-critical and already retracts the "2π = Gibbons–Hawking" reading for its density branch. The disagreement is a *fork*: APDM's headline adopts the density branch (superfluid carrier), USC prefers the rate branch (emergent inertia, ECCG dark matter). The two agree at $a_0(0) = cH_0/2\pi$ and diverge only in the sign of $a_0(z)$. This paper reports that high-z data, taken at face value through the MUSE chain, favour the USC (rate) fork and disfavour the APDM (density) fork — a discrimination *between two members of the unified family*, decided (conditionally) by the sign at z ~ 1.

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9. REPRODUCIBILITY

Every quantitative claim reproduces from a released script in https://github.com/spsingularity/apdm-galactic (a tagged release is archived at Zenodo, DOI 10.5281/zenodo.21525537):

- `branch_discrimination_test.py` (E-1): the two-branch templates over $z = 0.33 \to 1.44$; density $+16.7\sigma$, rate $+4.4\sigma$; the MUSE-preferred power $a_0 \propto$ H^{1.23}. Data: `desi_template_quantiles.csv`, `muse_release_galaxies.csv`, MUSE-DARK III published $a_0(z) = (1.00 \pm 0.04) + (1.59 \pm 0.11)z$.
- `d3prime_emergent_mond.py` (D-3′): the exact gated-Verlinde identity; the elastic/activated/density sub-branch exclusions (19.0σ / 27.3σ / 19.5σ); the KMS-tilt RAR pass at 0.057 dex.
- `d3prime_thermal_consistency.py`: the ~3×10^{-22} galactic bath-mode count that forces the vacuum framing.
- `d4prime_worldline_kernel.py` (D-4′): the master integral (10^{-16}, Gradshteyn 3.512); the kernel $(a/4\pi)\cot(\pi a/\kappa)$; the zero at $a = H/\sqrt{3}$; the deep-a and large-a limits; the dispersive/dissipative split.
- `mass_controlled_arot_trend.py` (E-1′): the raw-catalogue radius degeneracy (corr(r_max, z) = 0.87) that makes the released kinematics inconclusive.

The pre-registered falsifier matrix (P14–P18) is frozen at Zenodo `10.5281/zenodo.21415326`; nothing in it may be edited after freeze, and it predates the deciding data (JWST high-z kinematics, E-1″).

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10. COMPANION PAPERS AND REFERENCES

- S. Pandev, *USC program map* (companion note).
- S. Pandev, *Unified Structural-Entropy Cosmogenesis: a falsifiable framework* (Paper V).
- S. Pandev, *Accelerated-Phonon Dark Matter* (the APDM corpus; density-branch $a_0 \propto \sqrt{\rho_{DE}}$).

- *APDM — cross-check notes from the USC unification effort.*
- Pre-registration freeze: Zenodo `10.5281/zenodo.21415326`.
- Milgrom (1983, 1999, 2009); Verlinde (2016); Deser & Levin (1997); McGaugh, Lelli & Schombert (2016); MUSE-DARK III (high-z rotation).

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